DEVOPS

GIT (27/02/2026)

git => install

devops tools are open source tool => linux based tools

wsl => windows subsystem linux

wsl helps linux based tools to run on windows

Open git bash

Check git version

```
HP@DESKTOP-A1A0V5H MINGW64 ~
$ git --version
git version 2.51.1.windows.1
```

Create a folder (directory)

```
HP@DESKTOP-A1A0V5H MINGW64 ~
$ mkdir devops
```

Open the directory

```
HP@DESKTOP-A1A0V5H MINGW64 ~
$ cd devops
```

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops
$ git init
Initialized empty Git repository in C:/Users/HP/devops/.git/
```

Create file inside folder

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ vi demo.py
```

vi file name
I => insert
Type content

Esc
:wq
Enter

Use cat to view content inside file

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ cat demo.py
a=10
b=10
c=a+b
print(c)
```

Use git cmd to add the file inside git

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git add demo.py
warning: in the working copy of 'demo.py', LF will be replaced by CRLF the next time Git touches it
```

git status is used to check it status

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   demo.py


HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$
```

Now the file is save but not committed need to be commit

This command is used to commit the file inside git

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git commit -m "test commit"
[master (root-commit) 2e2ba91] test commit
 1 file changed, 4 insertions(+)
 create mode 100644 demo.py
```

The file is commited

Now there is no file to commit

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git status
On branch master
nothing to commit, working tree clean
```

Making changes in file and not added

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   demo.py

no changes added to commit (use "git add" and/or "git commit -a")
```

Without adding can’t commit

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git commit -m "test commit"
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   demo.py

no changes added to commit (use "git add" and/or "git commit -a")
```

After adding we can commit the file

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git add demo.py
warning: in the working copy of 'demo.py', LF will be replaced by CRLF the next time Git touches it

HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git commit -m "test commit"
[master 587e187] test commit
 1 file changed, 1 insertion(+), 1 deletion(-)

HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
```

This cmd is used to connect the git and github

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git config --global user.email "lekhapandic@gmail.com"

HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
```

Push pull perform panna namma enda repository la pandrom nu sollanum so create a repo in github

git-dev created

git remote cmd used to push the local repository to remote repository

...or push an existing repository from the command line

```
git remote add origin https://github.com/Lekha15-cse/git-dev.git
git branch -M main
```

Give link between quotes

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git remote add origin "https://github.com/Lekha15-cse/git-dev.git"
```

branch => branch name
Eg: one project diff branch like frontend and backend so frontend pandradu backend ku teriyatha same like backend pandradu frontend ku teriyatu

Branching strategy: to avoid clashes

Github => main branch (default)
Git => local branch (default)

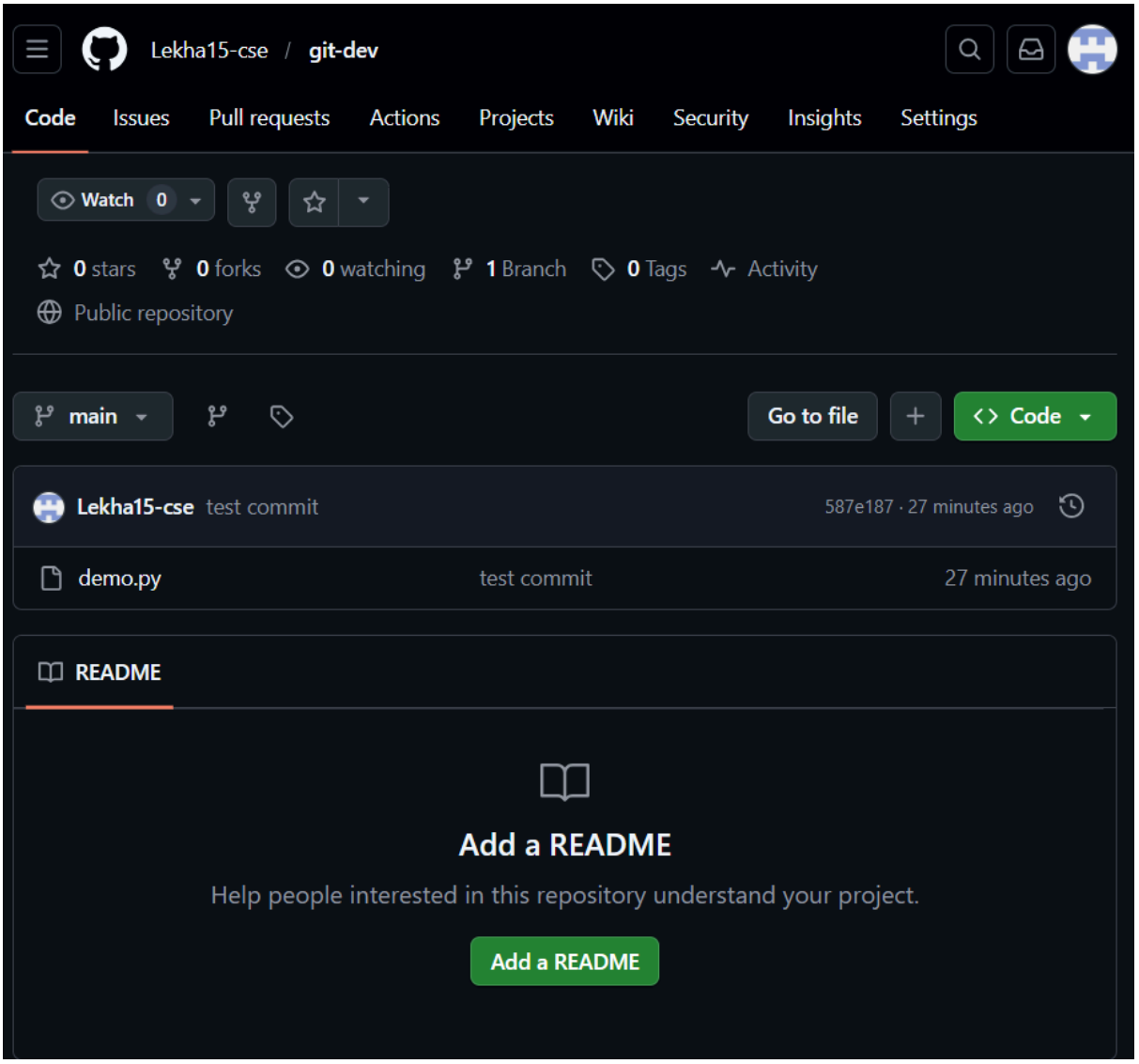
```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (master)
$ git branch -M main
```

Push the file into github

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (main)
$ git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 473 bytes | 118.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Lekha15-cse/git-dev.git
 * [new branch]      main -> main

HP@DESKTOP-A1A0V5H MINGW64 ~/devops (main)
```

Checking the file is pushed in github



Suppose for few days github we didn't use then key will be expired so the push cmd will won't work at that time we need to create a key

```
HP@DESKTOP-A1AOV5H MINGW64 ~/devops (main)
$ ssh-keygen
```

This cmd is used to generate a key

After key generate copy it and (public key generated nu oru path varum atha copy panniko)

Clip < (copied path paste here)
This clip cmd is used to copy the key file (for checking paste panni paru)

Go to github (right corner profile click panniko)

Move to settings
SSH and GPG keys (tap it in left panel)

SSH keys (click new SSH keys)

Title (give a title for key)
Key type (authenticate type)
Key (paste pannu copy panna key)
Save ssh key
Email verification

Now the added key will be showed

Already pushed means everything updated will come otherwise it will push

```
HP@DESKTOP-A1AOV5H MINGW64 ~/devops (main)
$ git push origin main
Everything up-to-date
```

git add . Used to add all files

```
HP@DESKTOP-A1A0V5H MINGW64 ~/devops (main)
$ git add .
warning: in the working copy of 'sec.py', LF will be replaced by CRLF the next time Git touches it
```